Consolidated Statement Errata

Chen–Ruau–Shen chemotaxis trilogy

Internal research note for the `ShenWork` Lean 4 formalization

15 July 2026

Abstract

This note consolidates the *statement-level* discrepancies found while formalizing the three papers in Lean 4. Every item is backed by a machine-checked Lean theorem that is `sorry`-free and depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`. Items are separated into (A) genuine paper-statement issues worth the authors' attention, and (B) excluded items — Lean-formalization artifacts that are *not* paper errors, listed for transparency. Full technical write-ups live in the per-theorem files `working_notes/paper{1,2,3}_*_amendment`; this note is the index and summary.

Papers and conventions

TW

"Traveling waves for repulsion/attraction chemotaxis with logistic source" (arXiv:2605.04401); repo `ShenWork/Paper1/`.

Part I

"...with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence" (arXiv:2512.14858); repo `ShenWork/Paper2/`.

Part II

"..., II: Persistence and stabilization" (arXiv:2604.02599); repo `ShenWork/Paper3/`.

Severity legend. **GENUINE-FALSE** — the printed conclusion is false on an admissible parameter regime (a hypothesis must be added). STATEMENT-FIX — the conclusion is true once a stated condition is added or the wording corrected (the printed form is over-stated, ill-posed, or has a proof gap, but is repairable).

1 Genuine paper-statement issues

A1 — TW, Theorems 1.2 & 1.3 (nonlinear orbital stability): the stability weight window is over-stated. Statement-Fix (with a genuine local proof gap)

Location

Theorem 1.2 statement and eqs. (1.21)/(1.22); Theorem 1.3 tail hypothesis $\kappa_1 > \kappa$. Proof steps (5.18),(5.19),(5.23),(5.27), (5.29)–(5.33),(5.35); Lemmas 5.2/5.3.

What is wrong

Seven independent defects in the §5 energy argument. The load-bearing ones:

1. **(5.35) root comparison is in the wrong direction.** With $q(\kappa) = A\kappa + B > 0$, κ is *not* between the two roots of q; it lies strictly below the lower perturbed root. The energy computation therefore does not establish decay for all $\eta > \kappa$ — the printed window $\kappa < \eta$ is over-stated.
2. **(1.21) uses the wrong spatial coordinate:** it prints the laboratory-frame weight $e^{2\eta x}$, but §5 estimates the *co-moving* weight; the two differ by $e^{2\eta ct}$.
3. **(5.35) exponential sign:** for $\lambda < 0$ the text writes $e^{-\lambda t} \to 0$, but $e^{-\lambda t}$ grows; the decay factor should be $e^{\lambda t}$.
4. **(5.18)/(5.19):** two chemotaxis sign errors (the $a_{m+\gamma}$ term and the $b_4 z$ term).
5. **(5.27) J_1 Young term drops a b_1:** should be b_1^2, not b_1; on the negative-χ branch $|\chi|$ may be arbitrarily large, so the replacement is not an upper bound.
6. **(5.29)–(5.30) drop an $M^{2(\gamma-1)}$ resolver factor:** on the positive-χ branch M_χ can exceed 1, so that power is not identically 1.
7. **Lemma 5.2 is one-sided but (5.23) uses a two-sided bound:** the lemma gives $U_x/U \le C$, while (5.23) uses $|U_x/U| \le C$; for a decreasing wave $U_x/U \le 0$ the negative part is uncontrolled.

Correction

Use the corrected budget A_χ, B_χ (carrying $|\chi|^2 B_1^2/2$ and a common resolver factor K); the proven weight window is the narrower $\kappa_- < \eta < 1/(1 + |\chi|^{1/6})$; state (1.21) in co-moving form; Theorem 1.3's tail exponent should require $\kappa_1 > \kappa_-$.

Lean evidence

`Theorem12RootObstruction.lean` (`paper531_kappa_not_between_perturbed_roots`, `paper531_kappa_lt_rootMinus`, `paper531_positive_inside_stated_weight_window`, `paper531_printed_decay_factor_tendsto_atTop`, `paper531_corrected_decay_factor_tendsto_zero`); `Theorem12CoordinateAudit.lean`; `Theorem12MeanCoefficients.lean`; `Theorem12WeightedEnergy.lean`; `Theorem12LogDerivative.lean`. Full note: `paper1_theorem12_statement_amendment`.

Severity

Mixed. Defects 2–6 are STATEMENT-FIX. Defects **1+7** leave a genuine gap: the printed window $\kappa < \eta$ is not proven by §5 on a sliver just above κ (not a counterexample, but a real gap absent a new localized-coercivity argument). The corrected narrower window is true.

A2 — Part I, Theorem 1.2: missing $\neg(a > 0 \wedge b = 0)$ guard; the mixed branch is false. GENUINE-FALSE

Location

Theorem 1.2(1) and (1.2)(2); Remark 1.4(2). Hypotheses read only $a, b \ge 0,\ \beta \ge 1$, which admits the mixed branch $a > 0,\ b = 0$.

Counterexample

The constant-in-space solution $u(t,x) = c\,e^{at}$, $v = (\nu/\mu)u^\gamma$, has all spatial derivatives and the chemotaxis divergence equal to zero and solves $u_t = au$; it is a global positive classical solution with $\|u(t)\|_\infty = c\,e^{at} \to \infty$. Equivalently the Neumann mass identity gives $M'(t) = aM(t)$, forcing exponential growth when $a > 0$. So (1) is false on this branch; (2) is already false at $m = 1, \beta = 1, \chi_0 = 0$. The §4.2/4.3 proof relies on Proposition 2.4's mass bound, which only covers $a = b = 0$ or $a, b > 0$.

Correction

Replace the hypothesis by $(a = 0) \vee (b > 0)$ (exclude $a > 0 \wedge b = 0$); or, conservatively, $(a = b = 0) \vee (a > 0 \wedge b > 0)$. Remark 1.4(2) should say both parameters may vanish *together*, not arbitrary mixed nonnegatives.

Lean evidence

`IntervalDomainTheorem12Refutation.lean` (`not_Theorem_1_2_intervalDomain_when_a_pos_b_zero`). Verified: $\neg$ Theorem_1_2 on a concrete parameter set. Full note: `paper2_theorem12_statement_amendment`.

Severity

GENUINE-FALSE on the $a > 0$, $b = 0$ regime.

A3 — Part I, Theorem 1.3, alternative (iv): missing exponent-domain condition. Statement-Fix (proof gap)

Location

Theorem 1.3 alternative (iv) ($\beta \geq 1/2,\ \alpha = 2m + \gamma - 2$); §5.4 invokes Proposition 2.2.

What is wrong

§5.4 needs $s(P) = (P + \alpha)/\gamma > 1$, i.e. $P > 2 - 2m$ at the critical identity; but the seed exponent $q_* = \max\{1, N\alpha/2\}$ chosen in the proof does not guarantee $q_* > 2 - 2m$.

Counterexample (parameter wedge)

$N = 1,\ m = 1/4,\ \gamma = 7/2,\ \alpha = 2,\ \beta = 1$: then $\alpha = 2m + \gamma - 2$, $q_* = 1$, $s(q_*) = 6/7 < 1$, and the first disjunct of (1.25) holds automatically (no χ_0 smallness), yet Proposition 2.2 needs $P > 3/2$. The written proof does not reach these parameters.

Correction

Add to (iv) the condition $\max\{1, N\alpha/2\} > 2 - 2m$ (in 1-D: $\max\{1, \alpha/2\} > 2 - 2m$). Alternative (iii) ($\alpha = m + \gamma - 1$) does not have this gap.

Lean evidence

`IntervalDomainTheorem13Critical{Constants,Seed,Threshold,Bootstrap}.lean` (`boundedBefore_critical_case_iv_corrected`). Full note: `paper2_theorem13_case_iv_amendment`.

Severity

Statement-Fix — a proof gap, not a counterexample to the conclusion; holds after adding the condition.

A4 — Part I, Proposition 1.1: the finite-horizon alternative should hang on a maximal continuation. Statement-Fix (wording)

Location

Proposition 1.1's continuation / maximal-time alternative (cf. (1.14)/(1.15)).

What is wrong

Read literally as "for every finite local horizon $T_{\max}$, the maximal-time alternative (finite-time blow-up or decay-to-zero) holds," the statement is false: the positive logistic equilibrium is a classical solution on every finite horizon yet neither blows up nor decays. The intended content is the existence of a *distinguished maximal continuation*, not a dichotomy for arbitrary finite horizons.

Counterexample

The constant equilibrium $c = (a/b)^{1/\alpha}$ $(a, b > 0)$ refutes "every finite local witness satisfies the maximal-time alternative," already on the unit horizon.

Correction

State the alternative on the maximal-continuation carrier: either a finite branch carrying (1.14)/(1.15), or a global branch; the finite-$T_{\max}$ dichotomy appears only when the reachable horizon is bounded.

Lean evidence

`IntervalDomainCorrectedProposition11.lean` (`not_legacyFiniteHorizonAlternativeProducer_of_positive_equilibrium`; corrected `correctedProposition_1_1_of_standardContinuation_and_gluing`). The faithful realization is now the headline `correctedProposition_1_1_intervalDomainM` in `IntervalDomainMMaximalContinuationAlternative.lean`: it states the alternative as an *endpoint-tail* dichotomy — for every threshold M and every $S < T_{\max}$, the tail $(S, T_{\max})$ still carries a witness above M (`UpperEndpointTail`) or below any floor (`FloorEndpointTail`), i.e. the genuine $t \uparrow T_{\max}$ blow-up/decay of (1.14)/(1.15), realized exactly at the sup reachable horizon. The literal "some finite $t < T_{\max}$" form survives only inside a forgetful compatibility adapter and never enters the headline. Unconditional (takes only the parameter record), non-vacuous ($u_0 \equiv 1$ witness), sorry/axiom-free.

Severity

Statement-Fix — a wording over-statement, now resolved in the formalization; offered to the authors to decide whether the printed phrasing needs clarifying.

A5 — Part II, Theorem 2.1(1) (uniform persistence): missing the $a = 0 < b$ pure-decay regime; the conclusion is false there. GENUINE-FALSE

Location

Theorem 2.1 part (1), §4.1 proof. Hypothesis reads only $m \geq 1$ and asserts $\liminf_{t\to\infty} \inf_x u > 0$, but §4.1 splits only into $a = b = 0$ and $a > 0 \wedge b > 0$, omitting the admissible $a = 0 < b$.

Counterexample

With $a = 0,\ b > 0$, the constant-in-space solution $u(t, x) = (c^{-\alpha} + \alpha bt)^{-1/\alpha}$, $v = (\nu/\mu)u^\gamma$, solves $u_t = -bu^{1+\alpha}$; it is global, positive, and bounded by c, yet $\liminf_x u = 0$. Simplest instance $\alpha = \gamma = m = \mu = \nu = b = 1,\ a = \chi_0 = 0$: $u = v = 1/(1 + t)$.

Correction

Add to part (1) the split actually used in §4.1: $(a = b = 0) \vee (a > 0 \wedge b > 0)$. The remaining $a > 0,\ b = 0$ branch makes the persistence hypothesis vacuous (mass $M' = aM$ grows exponentially, so no global bounded solution exists), so no third branch is needed.

Lean evidence

`IntervalDomainPersistencePart1StatementObstruction.lean` (`not_Theorem_2_1_part1_intervalDomain_pureDecay`; corrected `Theorem_2_1_part1_corrected`). Verified: ¬ Theorem_2_1_part1 with the explicit decaying orbit. Full note: `paper3_theorem21_statement_amendment`.

Severity

GENUINE-FALSE on the $a = 0 < b$ regime.

A6 — Part II, Theorem 2.2(1) (linear stability/instability): the all-time C^1 estimate (2.12) is over-stated. Statement-Fix

Location

Theorem 2.2 part (1), third assertion — printed (2.12) — which claims the C^1 exponential estimate for all $t \geq 0$ (including $t = 0$) while the initial hypothesis (1.8) only requires $u_0 \in C(\overline{\Omega})$ positive.

What is wrong

Smallness is in L^∞ but the conclusion is in C^1 and demands $t = 0$; the two are unrelated at $t = 0$. u_0 may have $\|u_0 - u^*\|_{C^1} = \infty$, making the $t = 0$ instance meaningless; even for C^1 data, the C^1 norm inside an L^∞-ball can be arbitrarily large, so no single constant C controls $t = 0$.

Counterexample

On $(0, 1)$ with Neumann BC, $u_{0,N} = u^* + N^{-1/2}\cos(N\pi x)$: $\|u_{0,N} - u^*\|_\infty = N^{-1/2} \to 0$ (mass exactly u^* in the minimal model) but $\|u_{0,N} - u^*\|_{C^1} = N^{1/2}\pi \to \infty$.

Correction

Split the single printed assertion into the two the proof actually gives: **(A) strong-norm, all-time** — X^α-small ($\alpha \in (3/4, 1)$) $\Rightarrow$ all-time exponential decay in X^α (Henry sectorial theory); **(B) L^∞-small, eventual** — there is a data-dependent $t_0(u_0) > 0$ such that C^1 exponential decay holds for $t \geq t_0$. The linear stability/instability dichotomy itself is unaffected and correct.

Side note

The sufficient stability bound $\kappa < (\sqrt{\mu} + \sqrt{a\alpha})^2$ is a continuous infimum; on a bounded Neumann domain the spectrum is discrete, so the sharp stability boundary should be governed by the discrete infimum $\min_{n \geq 1} \sigma_n < 0$ (the continuous bound is only sufficient and may be strictly stronger).

Lean evidence

`Statements.lean` (`not_SectorialLocalExponentialRaw_constant_c1Distance`); `IntervalDomainSectorialCorrectedObstruction.lean`. Full note: `paper3_theorem22_statement_amendment`.

Severity

Statement-Fix — true after splitting into A+B; only the printed (2.12) form is not well-posed.

A7 — Part II, Theorem 2.5 (minimal model $a = b = 0$ stability): the same all-time C^1 over-statement. Statement-Fix

Location

Theorem 2.5: one pair of exponential constants is quantified before all bounded positive global solutions, and the C^1 estimate is required for all $t \geq 0$.

What is wrong

Same all-time $t = 0$ C^1 issue as A6, in the minimal-model + mass-constrained version: L^∞/mass-small data cannot give a uniform C^1 bound at $t = 0$.

Correction

State it in eventual form: orbit-dependent constants + an orbit-dependent entry time t_0.

Lean evidence

`IntervalDomainSectorialCorrectedObstruction.lean` (`not_intervalDomain_Theorem_2_5_original_allTime`); corrected `intervalDomain_Theorem_2_5_EventualGlobalStabilityFormula`. (The Lean refutation also layers in a v_0-anchoring interface detail — a Lean-interface matter, see B5 — but the all-time C^1 over-statement itself is at the paper level.)

Severity

Statement-Fix — true in eventual form.

2 Summary

#	Paper	Statement	Issue / severity
A1	TW	Thm 1.2 / 1.3	weight window $\kappa < \eta$ over-stated (7 defects; 2 leave a real gap) — Statement-Fi
A2	Part I	Thm 1.2	missing $\neg(a > 0 \wedge b = 0)$; false there — **GENUINE-FALSE**
A3	Part I	Thm 1.3(iv)	missing $\max\{1, N\alpha/2\} > 2 - 2m$ — Statement-Fix
A4	Part I	Prop 1.1	alternative should hang on maximal continuation — Statement-Fix
A5	Part II	Thm 2.1(1)	missing $a = 0 < b$ regime; false there — **GENUINE-FALSE**
A6	Part II	Thm 2.2(1)	all-time C^1 (2.12) over-stated — Statement-Fix
A7	Part II	Thm 2.5	same all-time C^1 over-statement — Statement-Fix

The two **GENUINE-FALSE** items (A2, A5) have been verified against explicit ¬[theorem] refutations with concrete counterexample orbits. The remaining five are Statement-Fix: the conclusion holds after the stated correction.

3 Excluded: Lean-formalization artifacts (not paper errors)

Listed so the authors know these were examined and ruled out as paper-level issues.

1. **TW `not_forall_Lemma_2_1`**: refutes an *abstract* semigroup-estimate structure with fake data; the paper's Lemma 2.1 is about the *concrete* heat semigroup. Scaffolding only.
2. **TW Lemmas 4.1 / 4.2 / Remark 4.2**: all use one degenerate trap-set profile. A formalization-strength issue (the barrier lemma holds for the actual strictly-positive constructed wave; the Lean version over-quantifies over the whole trap set), not a paper error.
3. **Part I Lemmas 2.1–2.4** (`one_not_bounded_by_exp_decay`): scalar facts showing the current Lean *undamped* heat-helper route cannot recover the $e^{-\delta t}$ damping and sharp $1 + t^{-1/2}$ factors by tuning constants. The paper's exponential decay comes from the equation's built-in damping (the $-u$ term) and is sharp and correct; the Lean route uses a bare undamped helper. Internal-route boundary, not a paper error.

4. **`not_paper2_theorem_1_1_implies_paper3_proposition_1_2`**: an artificial single-point abstract domain shows the finite-$T_{\max}$ bound of Part I Thm 1.1 does not imply Part II Prop 1.2's eventual boundedness *under the abstract API*. A proof-dependency remark, not a statement counterexample.
5. **Part II Thm 2.1(4) mass interface & Thm 2.2/2.5 v_0 re-anchoring**: `HasInitialMass`/`InitialTrace` constrain only the $t \to 0^+$ limit, not the stored zero-time slice. These are documented Lean-interface amendments (correct interface: `HasEquilibriumMassOnPositiveTimes`), not paper falsities.
6. **`PositiveInitialDatum` too weak**: Lean once wrote initial positivity as open-interior pointwise positivity (admitting $\inf = 0$), while the papers' (1.11)/(1.8) require the uniform floor $\inf_\Omega u_0 > 0$. The *paper hypothesis is correct*; the Lean predicate was too weak (now fixed). TW stated positivity correctly.
7. **TW Proposition 1.1 "Cauchy solution"**: Lean used strictly-positive data, while the paper quantifies over arbitrary *nonnegative* BUC data (including the zero solution); fixed. Lean definition was too strong; paper correct.
8. **Part II Thm 2.2(nonlinear) / 2.3 / 2.4 for $m > 1$**: the Lean closers currently gate $m = 1$, whereas the papers assert $m \geq 1$ and advertise extending the Lyapunov-functional method from $m = 1$ to $m > 1$. This is this formalization's not-yet-covered frontier, not a refutation of the papers' $m \geq 1$ claims.

Maintained as the project's consolidated statement-errata index. New statement-level findings should be appended here (with a `sorry`-free Lean witness) before forwarding to the authors.